HUMANOID AI ASSISTANT

Frontend Technical Stack & Architecture Overview

This document outlines the advanced, modular frontend architecture required to support a real-time, 3D humanoid AI assistant with OS-level integration.

1. CORE DESKTOP & WEB FRAMEWORK

Next.js + React

The backbone for component-based UI, routing, and high-performance rendering. Essential for complex AI dashboard states.

Electron

Provides native system access: file systems, multi-window overlays, and background processing for the AI assistant.

2. 3D AVATAR & VISUAL EXPERIENCE

Three.js / React Three Fiber / Drei

Powering the humanoid avatar with GPU-accelerated rendering. Supports R3F for seamless React integration and Drei for character animation helpers.

- **Framer Motion:** For smooth OS-like UI transitions and floating panels.
- **GSAP:** Cinematic effects such as holographic reveals and parallax motions.
- **Tailwind CSS:** Rapid, responsive styling using modern utility-first principles.

3. REAL-TIME INTERACTION LAYER

Voice & Audio

- **Deepgram:** Low-latency Speech-to-Text.

- **ElevenLabs:** Realistic AI voice output.

State & Streaming

- **Zustand:** Lightweight global AI memory.
- **Socket.IO:** Live bi-directional sync.
- **WebRTC:** Camera vision streaming.

4. KEY UI MODULES

- **AI Dashboard:** Task managers and memory timelines built with **shadcn/ui** and **Lucide** icons.
- **Live Vision:** Webcam input processing for "eyes" on the assistant.
- **System Tray Assistant:** Background AI agent accessible via the OS tray using Electron.